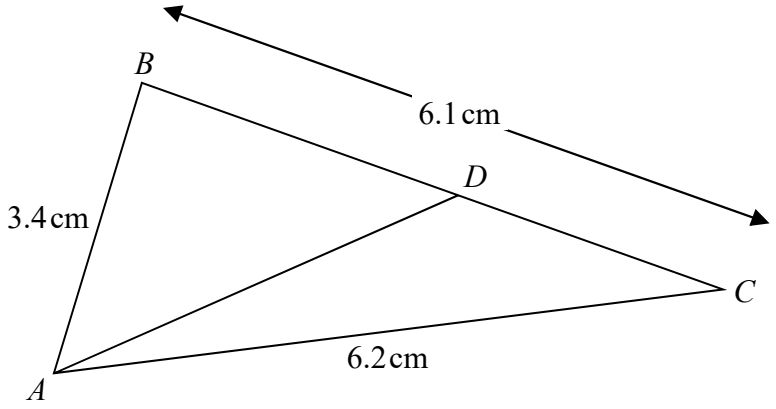
18 The diagram shows triangle ABC.



$AB = 3.4\,\text{cm}$ $AC = 6.2\,\text{cm}$ $BC = 6.1\,\text{cm}$

D is the point on BC such that

$$\text{size of angle } DAC = \frac{2}{5} \times \text{ size of angle } BCA$$

Calculate the length DC.
Give your answer correct to 3 significant figures.
You must show all your working.

Cosine rule: cos C = $(6.1^2+6.2^2-3.4^2)/(2\times6.1\times6.2)$.

So angle BCA ≈ 32.0805°.

Angle DAC = (2/5)×32.0805° ≈ 12.8322°.

Angle ADC = 180° − 32.0805° − 12.8322°

≈ 135.0873°. By sine rule,

DC/sin12.8322° = 6.2/sin135.0873°.

Therefore DC ≈ 1.95 cm (3 s.f.).

5 marks total
Cosine-rule angle
Angle DAC
Sine-rule setup
Complete DC calculation
1.95 cm

..... cm

(Total for Question 18 is 5 marks)

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